



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (CIVIL ENGINEERING)

ECV 205: CIVIL ENGINEERING MATERIALS I

DATE: Wednesday, 20th March 2019

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

This exam consists of FOUR questions

Question 1 is compulsory (30 marks)

Attempt any other 2 questions (20 marks each)

Question 1a (5 marks)

1. As a trainee engineer you are tasked with buying cement for making concrete. Although you bought cement from a reputable hardware shop, you realize that the quality of one of the bags had deteriorated and could not be used on site. Explain briefly what may have caused such deterioration.

Question 1b (15 marks)

Using DoE mix design procedure, design a mix if the target mean strength is 25 N/mm^2 at 28 days and the required slump is 10-30 mm. The following data is given:

Cement:	Ordinary Portland
Aggregate type	
Coarse:	Uncrushed, maximum size 10 mm
Fine:	Uncrushed, grading zone 4
Maximum w/c ratio:	0.60
Maximum cement content:	550 kg/m ³
Minimum cement content:	300 kg/m ³

Question 1c (10 marks)

Explain briefly how you would conduct a 'Life cycle assessment' of a concrete bridge.

- Objective
- assessment
- inventory
- conclusion

Question 2a (10 marks)

What are the main factors affecting workability of fresh concrete?

- water cement ratio
- aggregate
- size
- shape
- time

Question 2b (10 marks)

Explain the statement 'watering maintenance should be strengthened after concrete structures have been casted'.

Question 3

Question 3a (10 marks)

Corrosion of cement paste occurs because of chemical reactions between external and internal environments. What measures would you apply to prevent corrosion in concrete?

Question 3b (10 marks)

What do you understand by the term admixtures as used in concrete technology? Outline the main functions of concrete admixtures.

Question 4

Question 4a (10 marks)

What are the main factors affecting compressive strength of concrete?

Question 4b (8 marks)

Explain the term 'Life cycle assessment' with respect to building materials such as masonry

Table 2.7-2 Approximate compressive strengths (N/mm^2) of concrete mixes made with a free-water/cement ratio of 0.5 (after DoE [35])

Type of cement	Type of coarse aggregate	Compressive strengths (N/mm^2)			
		Age (days)			
		3	7	28	91
Ordinary Portland	Uncrushed	18	27	40	48
	Crushed	23	33	47	55
Rapid-hardening Portland	Uncrushed	25	34	46	53
	Crushed	30	40	53	60

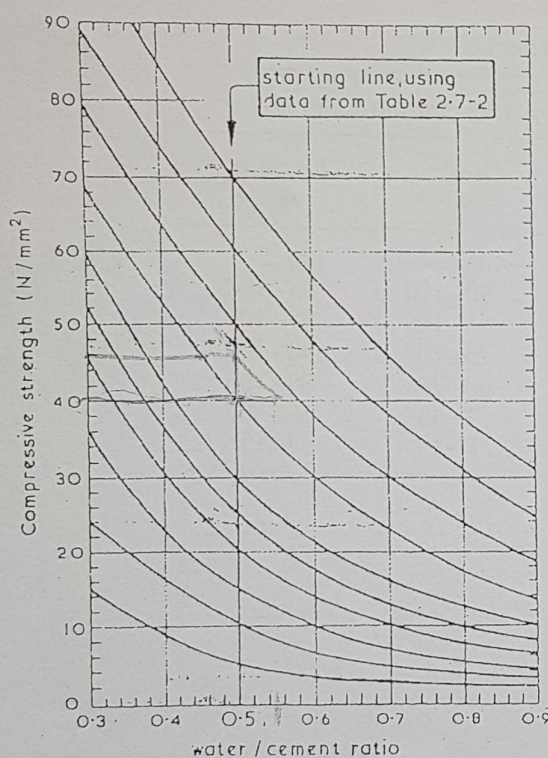


Fig. 2.7-2 Relationship between cube compressive strength and free-water/cement ratio (after DoE [35])

Table 2.7-3 Approximate free-water contents (kg/m^3) required to give various levels of workability (after DoE [35])

<i>Slump (mm):</i>		0-10	10-30	30-60	60-180
<i>VB time (seconds):</i>		2-12	12-6	6-3	3-0
<i>Maximum size of aggregate (mm)</i>	<i>Type of aggregate</i>				
10	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

Table 2.7-4 Grading limits for DoE mix design procedure

<i>Standard sieve</i>	<i>Percentage by weight passing standard sieves</i>			
	<i>Grading zone 1</i>	<i>Grading zone 2</i>	<i>Grading zone 3</i>	<i>Grading zone 4</i>
10 mm	100	100	100	100
5 mm	90-100	90-100	90-100	95-100
No. 7 (2.36 mm)	60-95	75-100	85-100	95-100
No. 14 (1.18 mm)	30-70	55-90	75-100	90-100
No. 25 (600 μm)	15-34	35-59	60-79	80-100
No. 52 (300 μm)	5-20	8-30	12-40	15-50
No. 100 (150 μm)	0-10	0-10	0-10	0-15

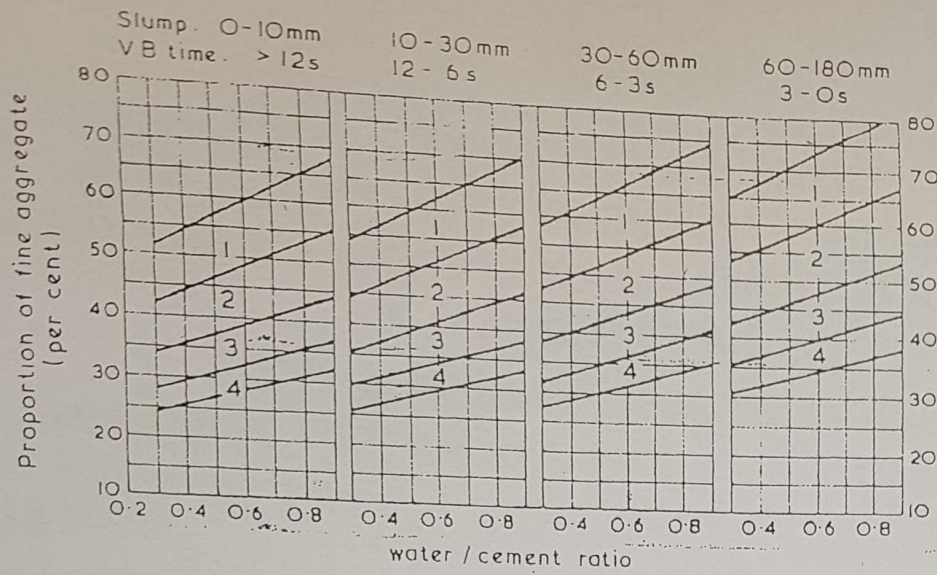


Fig. 2.7-3 Proportions of fine aggregate for grading zones 1, 2, 3, 4 (see Table 2.7-4)—for use with 10 mm nominal maximum size coarse aggregate (after DoE [35])



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FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN CIVIL ENGINEERING

ECV 205: CIVIL ENGINEERING MATERIALS I

DATE: WEDNESDAY 28TH FEBRUARY 2018

TIME: 2.00 P.M. - 4.00 P.M.

INSTRUCTIONS:

You are required to;

- Have an answer booklet
- Use the semi logarithmic graph paper provided for questions (4c) (Optional!)
- Have a non-programmable scientific calculator
- Answer QUESTION ONE and any TWO (2) questions.

QUESTION ONE (30 MARKS)

- (a) (i) Define the term "cement" as used in construction. *A binder - used in construction that binds aggregates and adheres to aggregates to bind them together* (1 mark)
- (ii) State the two raw materials used in the manufacture of Portland cement and name the chemical compound contained in each of the materials stated. *Argillaceous clay - calcium silicate
Calcareous material - calcium oxide found in limestone* (2 marks)
- (b) Distinguish between the following terms as they apply to cement; *Setting is changing of cement from fluid to solid state. Hardening is when cement and aggregates form a dense material.* (1 mark)
- (i) False set and flash set (1 mark)
- (c) (i) Define Pozzolana in accordance to American Standard ASTM C618. *Materials that react chemically with water to form void hardening cement.* (2 marks)
- (ii) In Pozzolanic cement, clinker is partly substituted with natural or artificial pozzolanic materials such as fly ash, silica fume, rice husk ash, etc. Give three (3) aim adding pozzolanic materials in cement. (3 marks)
- (d) (i) Define the term heat of hydration of cement, vary with time. Use an appropriate diagram. (4.5 marks)
- (e) State six main effects that aggregates grading has on concrete. (3 marks)

(d) With the aid of appropriate graphical sketch(s), explain the following types of aggregate grading patterns. (4.5 marks)

- (i) Gap grading - *Coarse aggregates have a uniform size but differ with a uniform size of fine aggregates*
- (ii) Uniform grading *The size of aggregates are about same size*
- (iii) Well graded grading *The size of aggregates increases in size from small to large*

(e) A split cylinder test was conducted on a 150 mm diameter x 300 mm long concrete cylinder in order to determine the tensile strength of concrete. The load at failure as read from the compression machine was 75 KN. Determine the tensile strength of this concrete. Comment on your computed value. (2 marks)

(f) State any four factors that affect the setting of concrete. (2 marks)

(g) Outline the objectives and concept of concrete mix design. (3 marks)

QUESTION TWO (20 MARKS)

(a) Give four reasons by aggregates are used in concrete production. (4 marks)

(b) The data shown in table 1.0 below, represents the mass (in grams) parameters found during experiment to determine the various specific gravities of coarse aggregates. (7 marks)

Table 1.0: Mass of coarse aggregates sample in various condition

Description	Mass (g)
Empty basket in air (W _a)	438.5
Weight of basket + Aggregates in air (W _b)	2438.5
Weight of basket + Aggregates in water	1597
Empty basket in water	390
Weight of tray + Aggregates oven dried	4419
Weight of empty tray	2468

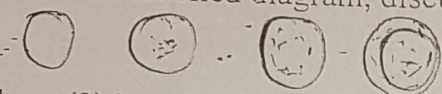
*Oven dry
Air dry
Surface saturated
dry
Saturated*

Determine the following specific gravities and water absorption of the aggregates. (7 marks)

(a) Specific gravity of saturated surface dry (SSD) condition

- (b) Absolute specific gravity
(c) Specific gravity on oven dry condition/apparent specific gravity
(d) Water absorption

(e) With the aid of a well labelled diagram, discuss the four moisture states that aggregates

exist in.  (6 marks)

(d) Explain the three (3) broad categories of deleterious substances that may be found in aggregates. Deleterious materials are those that have adverse effects on the aggregates and concrete mix. (3 marks)

1. Clay and friable materials
2. Soft materials
3. Low and low density materials

QUESTION THREE (20 MARKS)

(a) An on-going site development along Thika road concreting was done to columns, the Engineer advised the Contractor to prepare compressive cubes samples for 7 and 28 days test in the Laboratory. Describe test procedures to determine the compressive strength of the concrete. Why was the test necessary in this case? (5 marks)

(b) Define workability. State its significance in concrete mix. Workability is a property of concrete mix that affects ease at which concrete is mixed, placed, compacted and finished. (3 marks)

(c) While giving an example define and explain the influence/work of water-reducing agent in cement and concrete technology. Water reducing agent are materials that balance that reduce amount of water required in a concrete mix by 12 to 15% while maintaining the consistency and workability of concrete. (4.5 marks)

(d) Describe the three constituents of strengths that constitute to the overall strength of hardened concrete. Deformability, Durability, Compressive strength - compressive strength, shear bending (4.5 marks)

(e) What is "Bleeding" in concrete? Suggest ways of mitigating against bleeding in concrete works. Bleeding is the rising up of free water in a concrete mix to form a cement paste. (3 marks)

- Adding cement
- Reducing water content
- Adding admixtures

QUESTION FOUR (20 MARKS)

(a) List three advantages and three disadvantages that masonry construction have over other forms of construction. (6 marks)

(b) Explain the structural limitation of masonry structure. State measures undertaken in masonry construction to improve the structural limitations. (4 marks)

(c) Results for a sieve analysis conducted on a sample of fine aggregate for use in concrete production are summarized in the table 1.1 below. You are required to:

(i) Construct the grading curve of the aggregate samples and state two main advantages associated with good gradation. (7 marks)

- (ii) Determine the fineness modulus of the aggregates. State the significance of fineness modulus in the specification of fine aggregates. (3 marks)

Table 1.1: Sieve analysis test result for fine aggregates

Sieve size (mm)	Mass retained (g)
10	0
5	12
2.36	14
1.19	170.5
0.6	260.5
0.3	404.5
0.15	117.5
Pan	20.5

QUESTION FIVE (20 MARKS)

- (a) Define concrete mix design. State five (5) factors considered in mix design. (4.5 marks)
- (b) Using the DOE (department of environment) mix design procedure, design a concrete mix given the following information;

Minimum compressive strength at 28 days = 25 N/mm^2

Assumed margin = 5 N/mm^2 *1.64 1.96 2.58*

Specified slump range = 10 - 30 mm

Maximum W/C ratio = 0.60

Additional water added during mixing for enhanced workability = 28.5 kg/m^3

Cement type: Ordinary Portland cement (OPC)

Aggregates: Fine aggregates = uncrushed, Absorption capacity = 0.23%

Percentage of fine aggregates passing 0.6 mm sieve = 55%

Coarse aggregates = Crushed, Absorption Capacity = 0.91%

Combined specific gravities of the aggregates = 2.5

Maximum aggregate size = 20 mm

(15.5 marks)

APPENDICES

A-1 Extracts from BRE manual

Table 2 Approximate compressive strength of concrete mixes made with a free water/cement ratio of 0.5

Cement strength class	Type of coarse aggregates	Compressive strengths (N/mm ²)			
		Age (days)			
		3	7	28	91
OPC 42.5	Uncrushed	22	30	42	49
	Crushed	27	36	49	56
52.5	Uncrushed	29	37	48	54
	Crushed	34	43	55	61

Table 3 Approximate free-water contents (kg/m³) required to give various levels of workability

Slump (mm)	0-10	10-30	30-60	60-180
Vebe time (s)	>12	6-12	3-6	0-3
Max. aggregates size (mm)	Aggregates Type			
10	Uncrushed	150	180	205
	Crushed	180	205	230
20	Uncrushed	135	160	180
	Crushed	170	190	210
40	Uncrushed	115	140	160
	Crushed	155	175	190

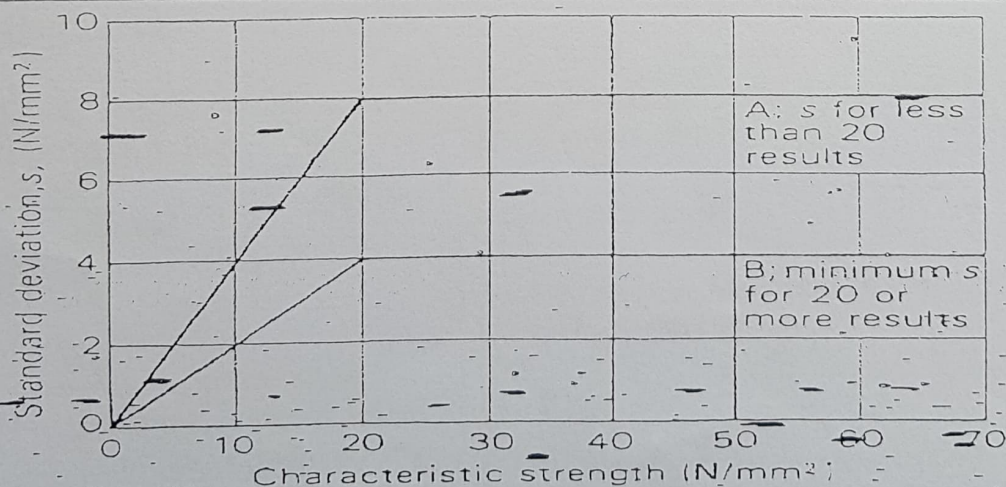


Figure 3 Relationship between standard deviation and characteristic strength

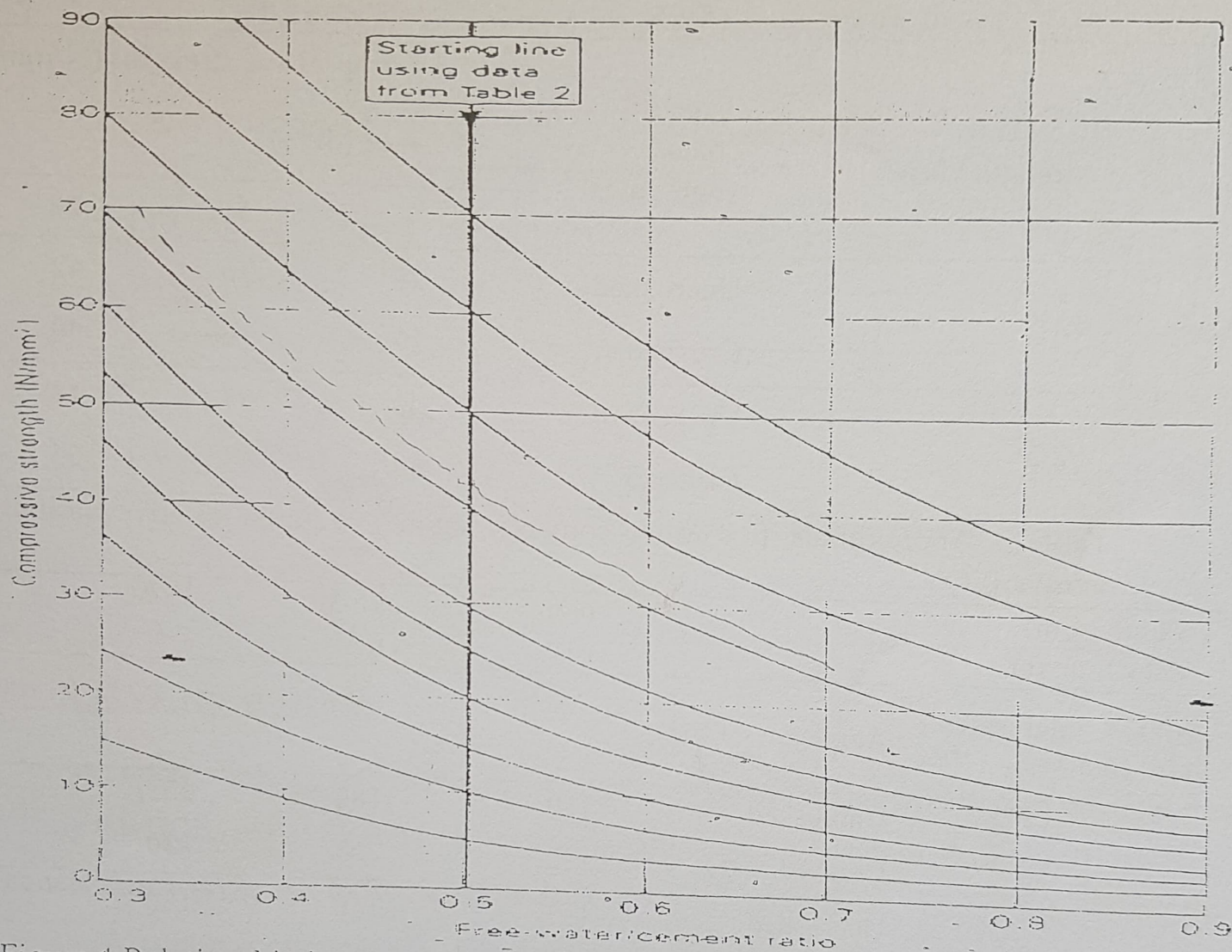


Figure 4 Relationship between compressive strength and free-water/cement ratio

Tutorial Questions

1. Why would an engineer want to classify building materials? What are different types
2. What are the main differences between Ordinary Portland cement and Rapid hardening Portland cement?
3. Outline various stages used in the manufacture of Ordinary Portland cement
4. Explain briefly the key properties required of Portland cement
5. By giving specific examples, explain why it may be necessary to apply some additives during manufacture of cement
6. As a student you are tasked with buying two bags of cement. Although you bought cement from the shop, you realized that the quality of one bag had deteriorated and could not be used for making concrete. Explain briefly what could have caused such deterioration.
7. a) Explain briefly the main requirements of a good concrete mix design
b) While studying a technical drawing, you come across a note: ' structural concrete to be class 1:2:4' what is your understanding of this statement. Outline the steps used in concrete mix design.
8. Outline various factors that may affect the setting and hardening of Portland cement
9. Explain the statement 'watering maintenance should be strengthened after concrete structures have been casted'.
10. What are the main technical properties of ordinary concrete
11. What are the main factors affecting workability of fresh concrete
12. Corrosion of cement paste occurs because of chemical reactions between external and internal environments. What measures would you apply to prevent corrosion in concrete
13. Outline the main functions of concrete admixtures
14. What are the main factors affecting compressive strength of concrete
15. Define the term life cycle assessment with respect to building materials such as masonry. How would you go about carrying out a life cycle analysis of a building project for housing engineering students? Explain by indicating the main stages of an environmental life cycle study.